

SMART BODA BODA AND BAJAJI QUEUE MANAGEMENT SYSTEM

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REG NO. 2022-04-08722

BSC.TELECOMMUNICATIONS ENGINEERING

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ABSTRACT

In many urban areas of Tanzania, boda boda and bajaji play a vital role as a fast and accessible means of public transport. However, most boda boda and bajaji stations lack an organized queue management system, leading to unfair service order, conflicts among riders, delays in customer service, and safety risks.

This project proposes a hardware based Smart Queue Management System that uses RFID technology, GSM communication, and display units to automatically manage rider queues at pickup points. Registered riders are detected at the station entrance using RFID readers and are automatically added to a queue. When a rider's turn arrives, the system notifies them through a GSM module and displays the currently serving rider on a public display. After serving, the rider exits through an outlet equipped with another RFID reader, allowing the system to update the queue and notify the next rider.

The proposed system aims to improve fairness, efficiency, and safety at boda boda and bajaji stations while minimizing human intervention.

PROBLEM STATEMENT

The absence of a proper queuing system at many bajaji and boda boda stations and pickup points has led to unfair service order and frequent conflicts among riders competing for customers. This situation often results in disorder, delays in customer service, enmity among riders, and in some cases physical conflict that may lead to injuries. Additionally, the lack of an organized serving mechanism creates confusion for customers and negatively affects overall service quality and public safety.

Therefore, addressing these challenges through an automated and well-structured queue management system is essential to improve fairness, efficiency in service delivery, and safety at boda boda and bajaji stations.

SOLUTION

To address these challenges, this project focuses on the design and implementation of a hardware based smart queue management system. The system automatically detects registered riders as they

arrive at the station using RFID technology, organizes them in a structured queue, and notifies them when their turn to serve arrives through a GSM module. A display unit shows the current rider being served, and the queue is automatically updated when a rider leaves the station after completing service. By implementing this system, fairness in service allocation is improved, conflicts among riders are minimized, waiting time is better managed, and overall efficiency at boda boda and bajaji stations is enhanced.

SYSTEM COMPONENT

Hardware

1. Two RFID readers , one placed at the entrance detects registered riders and other at the exit to detects the rider leaving the station
2. Arduino Mega 2560 to process the data, organizes riders into a queue, and controls system operations.
3. GSM module to send an SMS notification to the rider's registered mobile number
4. LED display shows the currently serving rider for public visibility.
5. EEPROM store permanently the rider information which include the rider name, phone number and plate number.

Software

Arduino IDE

METHOD USED (Step by step)

1. Establish system requirement

Functional requirements

- Detecting registered riders through RFID and adding them to the queue
- Sending GSM notifications and updating the queue.
- Adding, remove, show status and finding of rider information through SMS.

Non Functional requirement

- Offline
 - Reliable
2. Design the system
 3. Implement the system. The system implementation will involve assembling and integrating the required hardware components, including RFID readers, an Arduino microcontroller, a GSM module, an LED Display, EEPROM and a power supply, according to the proposed system design.

4. System testing. Testing will be conducted to verify correct detection of registered riders at entry and exit points, proper queue generation and updating, timely SMS notifications, and accurate display of queue information
5. Evaluation of the system and documentation

RECOMMENDATIONS FOR FUTURE WORK

To further improve the performance and applicability of the proposed system, the following recommendations are suggested:

1. Develop a centralized database to allow multiple boda boda and bajaji stations to share rider information and synchronize queue management.
2. Integrate a mobile application or web-based platform that allows riders and station administrators to monitor queue status in real time and receive notifications through the internet in addition to SMS.
3. Incorporate GPS technology to monitor rider location and automatically determine whether a rider is within the station before being added to the queue.
4. Add a backup power system, such as a rechargeable battery or solar power supply, to ensure continuous operation during power outages.
5. Integrate digital payment systems such as mobile money to support cashless service transactions where applicable.
6. Improve the user interface by using larger display screens or LED information boards that can easily be viewed by riders and passengers in crowded stations.

KEY FINDINGS

- The RFID readers successfully detected registered RFID cards and distinguished between authorized and unauthorized users. The entrance reader correctly added riders to the queue, while the exit reader removed riders after service completion. Multiple tests confirmed that queue positions were updated accurately after each rider entered or exited the station.
- The LCD displayed real-time information, including the current rider being served and the total number of riders waiting in the queue. The displayed information changed immediately whenever the queue status was updated.
- The external EEPROM successfully stored rider records permanently. Data remained intact after the system was powered off and restarted, confirming reliable non-volatile storage.
- The SIM800C GSM module successfully transmitted SMS notifications to riders when their turn arrived. Also the sms can be received from admin number for adding, removing find and show status of the rider in a queue.
- Overall, the hardware implementation demonstrated that the proposed Smart Queue Management system can provide a practical, fair, and efficient solution for managing boda boda and bajaji queues. The prototype achieved reliable rider identification, automatic queue management, real-time information display, permanent data storage, and SMS-based rider notification, making it suitable for deployment in organized transport stations.

APPENDICES

Github repo: <https://github.com/Shiru2024/Smart-boda-boda-and-bajaji-queue-management-system-.git>



Figure 1 Full Working prototype

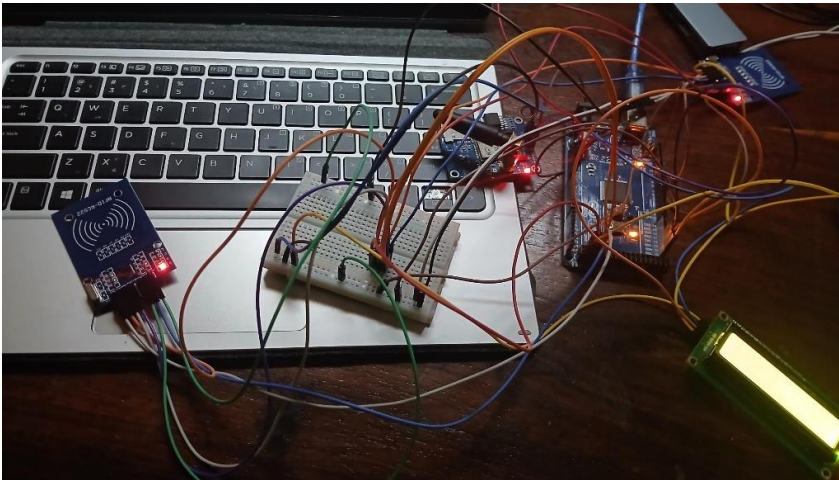


Figure 2 Circuit Layout



Figure 3 LCD showing driver queue

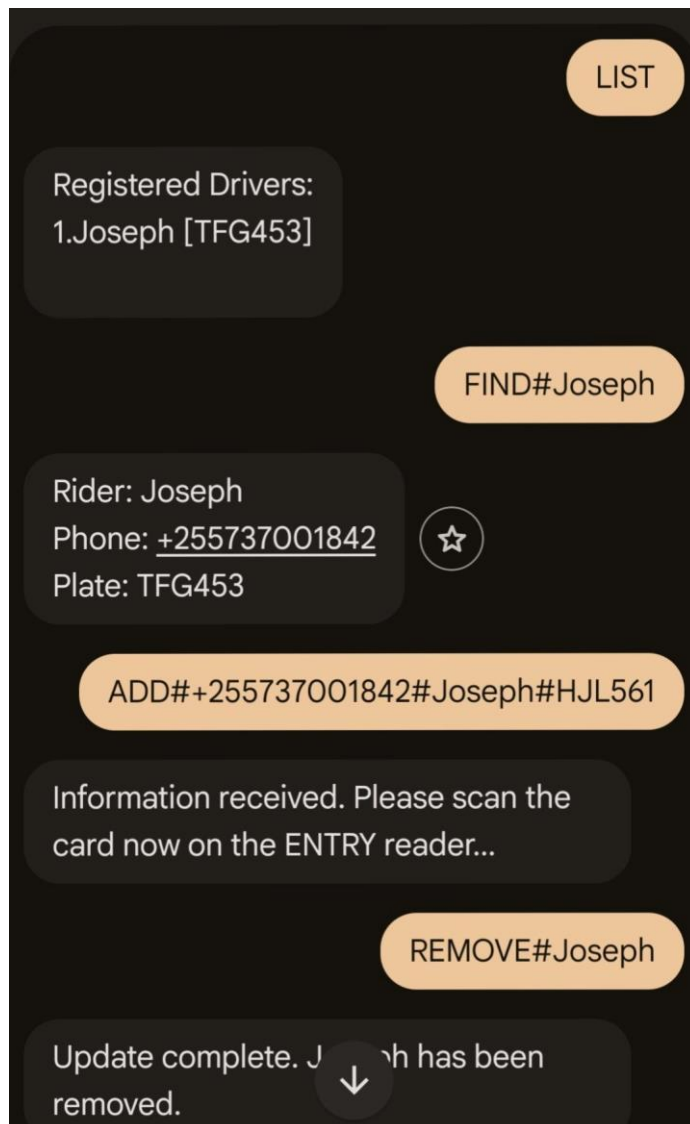


Figure 4 Admin CRUID functionalities



Figure 5 LCD Showing Rider Served Already

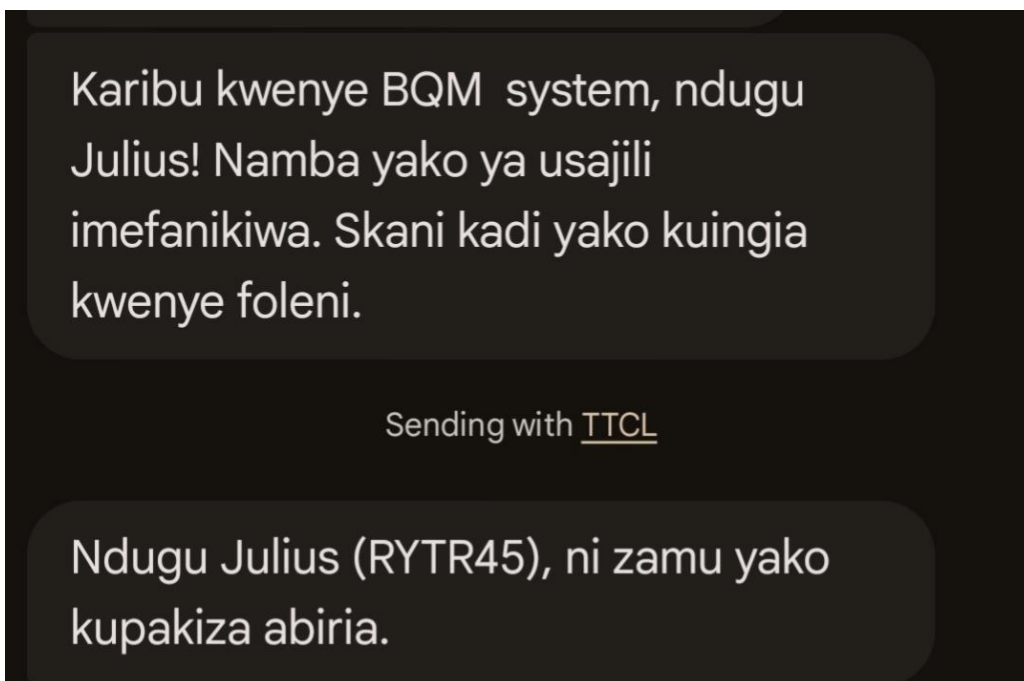


Figure 6 Rider SMS notification for serving